Regumetrica

Beta version

User manual

Version 1.0

www.regumetrica.com/um

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Contents

Introduction

Regumetrica simulates counterfactual revenue frames for electricity distribution networks in Sweden, based on a wide range of regulatory specifications. The current version computes revenue frames for the ongoing regulatory period, 2024–2027.

To get started, create an account at www.regumetrica.com/newuser. You will need a valid email address, which will also serve as your username [could we verify email addresses by sending a confirmation link?]. During registration, you will be asked to indicate your primary concession area of interest. Concession areas are the smallest geographical units for which revenue frames are determined. You can change your preferred concession area at any time. Several accounts may be connected to the same concession area. Each area has both a name and an identification number assigned by the regulator.

After logging in, you enter the main environment by opening a *case*. A case is a workspace that contains all assumptions for a specific regulatory simulation. Each case represents one set of assumptions and therefore produces one revenue frame. Cases must be named; by default, they are labeled consecutively ("Case 1," "Case 2," etc.), but you can also assign custom names. Initially, all regulatory assumptions are set according to the current regulatory model.

Once you have adjusted the assumptions to your preferred levels, the model is run. Simulations usually complete within a few seconds, though more complex configurations may take longer. Regumetrica then provides downloadable data detailing the revenue frame for the selected concession area, along with all underlying calculation components. In addition to data downloads, results can also be displayed graphically, for example as bar charts decomposing different components of the revenue frame. After running a model, you may choose to save the case. At present, each user can store up to 10 saved cases.

Model Components

The regulatory model is organized into *Base modules* and *Add-on modules*. Base modules represent the core components of the revenue frame calculation: Regulatory asset base valuation, Depreciation, Cost of capital, Operating expenditures, Efficiency incentive, and Model outputs.

Each base module contains *Parameters* and *Variables*:

- *Parameters* are fixed values that define the structural relationships within the regulatory model. They capture assumptions or policy choices that apply uniformly across regulated entities. For example, the regulatory *WACC* (Weighted Average Cost of Capital) determines the allowed rate of return on capital and is itself derived from a set of *sub-parameters*, such as the risk-free rate of return. Another example of a parameter is the *economic lifetime* assigned to a specific asset category, which determines the rate of capital depreciation over time.

- *Variables* are measurable inputs that vary across regulated entities. They correspond to real-world data, such as asset quantities, operating costs, or energy delivered, that the model uses to calculate revenue frames. Most variables are reported to the regulator through the KENT excel reporting template. Regumetrica users have the option of adjusting such variables manually, or to upload standardized KENT excel sheets.

Add-on modules extend the base regulatory model. They are designed to either explore regulatory scenarios that fall outside the scope of the current regulatory framework, or to facilitate the computation of revenue frames for specific strategic choices made by the regulated entity. An example of an add-on module exploring alternative regulatory scenarios is the *Benchmarking* module. This module can compute efficiency scores using the currently employed Data Envelopment Analysis (DEA) model. An example of an add-on module for understanding the consequences of strategic choices is the *Merger* module, in which revenue frames are computed conditional on that two or more concession areas are merged into one area. The *Merger* module builds on assumptions of economies of scale in operating expenditures derived from previous studies. Add-on modules are described in Section 7.

1 Regulatory asset base valuation

1.1 Parameters

Table 1: Regulatory asset base valuation

Description	Value	Parameter-ID
1.1 General asset valuation parameters		
General scaling factor for asset valuation[a]	1.00	1.1.1
1.2 Asset type specific valuation parameters		
Scaling factor for andra markarbeten och byggnader, linjekoncession	1.00	1.2.1
Scaling factor for annan ledning, linjekoncession	1.00	1.2.2
Scaling factor for annan ledning, områdeskoncession	1.00	1.2.3
Scaling factor for annan luftledning, linjekoncession	1.00	1.2.4
Scaling factor for it-system	1.00	1.2.5
Scaling factor for kabelskåp	1.00	1.2.6
Scaling factor for ledning med en spänning om 220 kv eller mer, med undantag för luftledning, linjekoncession	1.00	1.2.7
Scaling factor for luftledning med en spänning om 220 kv eller mer, linjekoncession	1.00	1.2.8
Scaling factor for luftledning, områdeskoncession	1.00	1.2.9
Scaling factor for markarbeten och byggnader med anknytning till ett ledningsnät med en spänning om 220 kv eller mer, linjekoncession	1.00	1.2.10
Scaling factor for markarbeten och byggnader, områdeskoncession	1.00	1.2.11
Scaling factor for mätare	1.00	1.2.12
Scaling factor for nätstation	1.00	1.2.13
Scaling factor for shuntreaktor	1.00	1.2.14
Scaling factor for styr- och kontrollutrustning	1.00	1.2.15
Scaling factor for ställverk utan sekundärapparater	1.00	1.2.16
Scaling factor for transformator	1.00	1.2.17

[a] Scaling factors are applied multiplicatively to asset norm values from Bilaga 6.

1.2 Variables

Table 2: Input variables: Regulatory asset base valuation

Description	Variable-ID
10.2–10.18 Asset quantities by type	
Andra markarbeten och byggnader, linjekoncession	10.2
Annan ledning, linjekoncession	10.3
Annan ledning, områdeskoncession	10.4
Annan luftledning, linjekoncession	10.5
It-system	10.6
Kabelskåp	10.7
Ledning med en spänning om 220 kV eller mer, med undantag för luftledning, linjekoncession	10.8
Luftledning med en spänning om 220 kV eller mer, linjekoncession	10.9
Luftledning, områdeskoncession	10.10
Markarbeten och byggnader med anknytning till ett ledningsnät med en spänning om 220 kV eller mer, linjekoncession	10.11
Markarbeten och byggnader, områdeskoncession	10.12
Mätare	10.13
Nätstation	10.14
Shuntreaktor	10.15
Styr- och kontrollutrustning	10.16
Ställverk utan sekundärapparater	10.17
Transformator	10.18

Asset quantities are sourced from the KENT reporting template. Units vary by asset type.

Table 3: Output variables: Regulatory asset base valuation

Description	**Variable-ID**
Total asset value for all asset types	11.1
11.2–11.18 Asset values by type	
Andra markarbeten och byggnader, linjekoncession	11.2
Annan ledning, linjekoncession	11.3
Annan ledning, områdeskoncession	11.4
Annan luftledning, linjekoncession	11.5
It-system	11.6
Kabelskåp	11.7
Ledning med en spänning om 220 kV eller mer, med undantag för luftledning, linjekoncession	11.8
Luftledning med en spänning om 220 kV eller mer, linjekoncession	11.9
Luftledning, områdeskoncession	11.10
Markarbeten och byggnader med anknytning till ett ledningsnät med en spänning om 220 kV eller mer, linjekoncession	11.11
Markarbeten och byggnader, områdeskoncession	11.12
Mätare	11.13
Nätstation	11.14
Shuntreaktor	11.15
Styr- och kontrollutrustning	11.16
Ställverk utan sekundärapparater	11.17
Transformator	11.18

Asset values in kr. Values represent assets before depreciation. Tail component values depend on the age of each specific component and cannot be adjusted directly; upload a modified KENT file to change tail values.

2 Depreciation

2.1 Parameters

Table 4: Ordinary and tail asset lifetimes

Description	Value		Parameter-ID
2.1 Asset lifetime parameters (years)	**Ordinary**[a]	**Tail**[b]	
Andra markarbeten och byggnader, linjekoncession	100	24	2.1.1; 2.1.2
Annan ledning, linjekoncession	100	24	2.2.1; 2.2.2
Annan ledning, områdeskoncession	100	24	2.3.1; 2.3.2
Annan luftledning, linjekoncession	100	24	2.4.1; 2.4.2
It-system	20	4	2.5.1; 2.5.2
Kabelskåp	60	14	2.6.1; 2.6.2
Ledning med en spänning om 220 kv eller mer, med undantag för luftledning, linjekoncession	80	20	2.7.1; 2.7.2
Luftledning med en spänning om 220 kv eller mer, linjekoncession	120	30	2.8.1; 2.8.2
Luftledning, områdeskoncession	80	20	2.9.1; 2.9.2
Markarbeten och byggnader med anknytning till ett ledningsnät med en spänning om 220 kv eller mer, linjekoncession	80	20	2.10.1; 2.10.2
Markarbeten och byggnader, områdeskoncession	100	24	2.11.1; 2.11.2
Mätare	20	4	2.12.1; 2.12.2
Nätstation	80	20	2.13.1; 2.13.2
Shuntreaktor	80	20	2.14.1; 2.14.2
Styr- och kontrollutrustning	30	6	2.15.1; 2.15.2
Ställverk utan sekundärapparater	80	20	2.16.1; 2.16.2
Transformator	100	24	2.17.1; 2.17.2

[a] Ordinary lifetime applies to assets that have not yet reached full depreciation.
[b] Tail lifetime applies to assets that remain in service beyond their ordinary economic lifetime.
Parameter-IDs ending with .1 refer to ordinary lifetime; those ending with .2 refer to tail lifetime.

2.2 Variables

Input variables: 10.X-10.X above, plus the existing tail components that are either existing or from KENT-data.

Table 5: Output variables: Depreciation cost

Description	**Variable-ID**
Total depreciation cost for all asset types	20.1.1; 20.1.2
20.2–20.18 Cost of depreciation by asset type (ordinary and tail)	
Andra markarbeten och byggnader, linjekoncession	20.2.1; 20.2.2
Annan ledning, linjekoncession	20.3.1; 20.3.2
Annan ledning, områdeskoncession	20.4.1; 20.4.2
Annan luftledning, linjekoncession	20.5.1; 20.5.2
It-system	20.6.1; 20.6.2
Kabelskåp	20.7.1; 20.7.2
Ledning med en spänning om 220 kV eller mer, med undantag för luftledning, linjekoncession	20.8.1; 20.8.2
Luftledning med en spänning om 220 kV eller mer, linjekoncession	20.9.1; 20.9.2
Luftledning, områdeskoncession	20.10.1; 20.10.2
Markarbeten och byggnader med anknytning till ett ledningsnät med en spänning om 220 kV eller mer, linjekoncession	20.11.1; 20.11.2
Markarbeten och byggnader, områdeskoncession	20.12.1; 20.12.2
Mätare	20.13.1; 20.13.2
Nätstation	20.14.1; 20.14.2
Shuntreaktor	20.15.1; 20.15.2
Styr- och kontrollutrustning	20.16.1; 20.16.2
Ställverk utan sekundärapparater	20.17.1; 20.17.2
Transformator	20.18.1; 20.18.2

All values in kr. Variable-IDs ending with .1 refer to ordinary depreciation; those ending with .2 refer to tail depreciation. Depreciation is calculated using linear depreciation over the asset lifetime.

3 Cost of capital

3.1 Parameters

Table 6: Weighted Average Cost of Capital (WACC)

Description	Value	Determined by	Parameter-ID
3.1 Base parameters			
Debt ratio[a]	0.36		3.1.1
Asset beta[b]	0.37		3.1.2
Risk-free rate[c]	0.0287		3.1.3
Market risk premium	0.0668		3.1.4
Credit risk premium	0.0114		3.1.5
Tax rate	0.206		3.1.6
Inflation, CPIF forecast	0.0202		3.1.7
3.2 Endogenously determined parameters			
Equity beta[d]	0.54	Asset beta, Debt ratio, Tax rate	3.2.1
Nominal cost of equity after tax	0.0645	Risk-free rate, Equity beta, Market risk premium	3.2.2
Nominal cost of debt after tax	0.0318	Risk-free rate, Credit risk premium, Tax rate	3.2.3
Nominal WACC before tax	0.0664	All parameters above	3.2.4
Real WACC before tax	0.0453	Nominal WACC before tax, Inflation	3.2.5

[a] Debt ratio g represents the share of debt in total capital.
[b] Asset beta β_A measures systematic risk of unlevered assets.
[c] Based on 10-year Swedish government bond yields.
[d] Equity beta computed as $\beta_E = \beta_A \cdot (1 + (1 - \tau) \cdot g/(1 - g))$.
When endogenously determined parameter values are altered, those changes override modifications to sub-parameters.

3.2 Variables

Input variables: 10.X-10.X above, plus the existing tail components that are either existing or from KENT-data.

Table 7: Output variables: Capital cost

Description	**Variable-ID**
Total capital cost for all asset types	30.1.1; 30.1.2
30.2–30.18 Capital cost by asset type (ordinary and tail)	
Andra markarbeten och byggnader, linjekoncession	30.2.1; 30.2.2
Annan ledning, linjekoncession	30.3.1; 30.3.2
Annan ledning, områdeskoncession	30.4.1; 30.4.2
Annan luftledning, linjekoncession	30.5.1; 30.5.2
It-system	30.6.1; 30.6.2
Kabelskåp	30.7.1; 30.7.2
Ledning med en spänning om 220 kV eller mer, med undantag för luftledning, linjekoncession	30.8.1; 30.8.2
Luftledning med en spänning om 220 kV eller mer, linjekoncession	30.9.1; 30.9.2
Luftledning, områdeskoncession	30.10.1; 30.10.2
Markarbeten och byggnader med anknytning till ett ledningsnät med en spänning om 220 kV eller mer, linjekoncession	30.11.1; 30.11.2
Markarbeten och byggnader, områdeskoncession	30.12.1; 30.12.2
Mätare	30.13.1; 30.13.2
Nätstation	30.14.1; 30.14.2
Shuntreaktor	30.15.1; 30.15.2
Styr- och kontrollutrustning	30.16.1; 30.16.2
Ställverk utan sekundärapparater	30.17.1; 30.17.2
Transformator	30.18.1; 30.18.2

All values in kr. Variable-IDs ending with .1 refer to ordinary capital cost; those ending with .2 refer to tail capital cost. Capital cost is computed as WACC $\times$ regulatory asset base.

3.3 Adjustment of cost of capital

Adjustment of the WACC are made with respect to network losses, utilization rate, and interruptions. Where norm values vary across firms, they are treated as variables. For a full description of these incentives, see Appendix X.

3.3.1 Parameters

Table 8: Cost of capital adjustment

Description	**Value**		**Parameter-ID**
3.3 Overall adjustment			
Max total adjustment (share of WACC)[a]	0.333		3.3.1
3.4 Network loss adjustment			
Loss incentive scaling parameter	1.00		3.4.1
Network loss incentive sharing factor[b]	0.75		3.4.2
Average network loss cost (kr/MWh)	734		3.4.3
3.5 Utilization rate adjustment			
Utilization rate incentive scaling parameter	1.00		3.5.1
3.6 Interruption adjustment			
Cost scaling parameter applied to ILE and ILEffekt	1.00		3.6.1
CPI 2024[c]	1.28		3.6.2
CPI 2025	1.31		3.6.3
CPI 2026	1.33		3.6.4
CPI 2027	1.36		3.6.5
CEMI4 correction factor[d]	0.25		3.6.6
	Unplanned	**Planned**	
Interruption energy cost – ILE (kr/kWh)[e]			
Household	5.84	4.98	3.6.7, 3.6.8
Agriculture	34.35	14.10	3.6.9, 3.6.10
Trade/Services	175.06	79.31	3.6.11, 3.6.12
Industry	159.96	76.00	3.6.13, 3.6.14
Public sector	96.97	43.70	3.6.15, 3.6.16
Boundary points	96.01	45.16	3.6.17, 3.6.18
Interruption power cost – ILEffekt (kr/kW)[e]			
Household	1.95	1.85	3.6.19, 3.6.20
Agriculture	9.78	1.72	3.6.21, 3.6.22
Trade/Services	17.78	5.94	3.6.23, 3.6.24
Industry	70.75	20.71	3.6.25, 3.6.26
Public sector	7.65	0.92	3.6.27, 3.6.28
Boundary points	22.18	7.08	3.6.29, 3.6.30

[a] Each sub-incentive is capped at ±1/3 of regulatory return; total adjustment also capped at 1/3.
[b] Share of network loss savings/costs passed to the network operator.
[c] Consumer price index relative to 2017 base year, used to inflate interruption costs.
[d] Maximum share of quality adjustment that can be reduced due to CEMI4 correction.
[e] Interruption costs in 2017 SEK, derived from willingness-to-pay studies.

3.3.2 Variables

Table 9: Input variables: Adjustment of cost of capital

Description	Variable-ID
30.2 Network loss adjustment	
Network loss norm level (Nf_{norm})	30.2.1
Network loss observed (Nf_{utfall})	30.2.2
Energy input (E_{in})	30.2.3
30.3 Utilization rate adjustment	
Utilization rate norm level (Ug_{norm})	30.3.1
Utilization rate observed (Ug_{utfall})	30.3.2
Cost for upstream network ($k_{\text{överl.nät mm}}$)	30.3.3
30.4 Interruption adjustment	
Customers Experiencing Multiple (4) Interruptions, norm ($CEMI4_{norm}$)	30.4.1
Customers Experiencing Multiple (4) Interruptions, observed ($CEMI4_{utfall}$)	30.4.2
Annual average power (ÅME^k)	
Household	30.4.3
Agriculture	30.4.4
Trade/Services	30.4.5
Industry	30.4.6
Public sector	30.4.7
Boundary points	30.4.8

	Norm		Observed	
	Unplanned	**Planned**	**Unplanned**	**Planned**
Average Interruption Time (AIT)				
Household	30.4.9	30.4.10	30.4.11	30.4.12
Agriculture	30.4.13	30.4.14	30.4.15	30.4.16
Trade/Services	30.4.17	30.4.18	30.4.19	30.4.20
Industry	30.4.21	30.4.22	30.4.23	30.4.24
Public sector	30.4.25	30.4.26	30.4.27	30.4.28
Boundary points	30.4.29	30.4.30	30.4.31	30.4.32
Average Interruption Frequency (AIF)				
Household	30.4.33	30.4.34	30.4.35	30.4.36
Agriculture	30.4.37	30.4.38	30.4.39	30.4.40
Trade/Services	30.4.41	30.4.42	30.4.43	30.4.44
Industry	30.4.45	30.4.46	30.4.47	30.4.48
Public sector	30.4.49	30.4.50	30.4.51	30.4.52
Boundary points	30.4.53	30.4.54	30.4.55	30.4.56

All variables vary by firm and year. Units: Nf and Ug are shares (0–1); E_{in} in MWh; $k_{\text{överl.nät mm}}$ in kr; $CEMI4$ is a share (0–1); ÅME^k in kW; AIT in hours; AIF in count.

Table 10: Output variables: Adjustment of cost of capital

Description	Variable-ID
30.2 Network loss adjustment	
Network loss adjustment before cap	30.2.4
Network loss adjustment after cap	30.2.5
30.3 Utilization rate adjustment	
Utilization rate incentive before cap	30.3.4
Utilization rate incentive after cap	30.3.5
30.4 Interruption adjustment	
Interruption adjustment before CEMI4 correction	30.4.57
Interruption adjustment after CEMI4 correction	30.4.58
Interruption adjustment after CEMI4 correction and cap	30.4.59
30.5 Total adjustment	
Total adjustment after individual caps but before aggregate cap	30.5.1
Total adjustment after aggregate cap	30.5.2

All variables vary by firm and year. Units: kr. Positive values indicate revenue increases; negative values indicate revenue decreases.

4 Operating expenditures

4.1 Parameters

Table 11: Operating expenditures

Description	Value	Parameter-ID
4.1 General scaling parameters		
Scaling factor adjustable OPEX[a]	1.00	4.1.1
Scaling factor flexibility services	1.00	4.1.2
Scaling factor non-adjustable OPEX[b]	1.00	4.1.3

[a] Adjustable OPEX is subject to efficiency requirements from DEA benchmarking.
[b] Non-adjustable OPEX includes costs outside the network operator's direct control.

4.2 Variables

Table 12: Input variables: OPEX

Description	Variable-ID
40.1 Adjustable OPEX	
Adjusted OPEX ($OPEX_p$)	40.1.1
Flexibility service cost	40.1.2
40.2 Non-adjustable OPEX	
Total non-adjustable costs (prognosis)	40.2.1

All values in kr. Variables vary by firm and year. Adjustable OPEX is based on 4-year historical averages (2018–2021); non-adjustable OPEX is based on forecasted values.

5 Efficiency incentive

5.1 Parameters

Table 13: Data Envelopment Analysis

Description	Value	Parameter-ID
5.1 Outlier identification		
Outlier threshold (IQRs above Q3)[a]	2.00	5.1.1
5.2 Efficiency requirement conversion		
Maximum efficiency potential cap[b]	0.30	5.2.1
Realization time (years)[c]	8	5.2.2
Customer sharing factor[d]	0.50	5.2.3
5.3 Efficiency requirement bounds		
Minimum annual efficiency requirement	0.0100	5.3.1
5.4 Cost base application		
Apply efficiency requirement on TOTEX	No (OPEX only)	5.4.1

[a] Firms with efficiency scores exceeding Q3 + 2×IQR are excluded as outliers.
[b] Efficiency potential is capped at 30% regardless of DEA score.
[c] Period over which efficiency improvements are expected to be realized.
[d] Share of efficiency gains passed through to customers vs. retained by operator.

5.2 Variables

Table 14: Input variables: Data Envelopment Analysis

Description	Variable-ID
50.1 Input cost variables	
Adjustable OPEX ($OPEX_p$)[a]	40.1.1
CAPEX ($CAPEX$)[a]	30.6.1
50.2 Exogenous environmental variables	
Number of subscriptions (*Abonnemang*)	50.2.1
Number of substations (*Nätstationer*)	50.2.2
Energy delivered, low voltage (E_{LS})	50.2.3
Energy delivered, high voltage (E_{HS})	50.2.4
Maximum power from upstream network (P_{max})	50.2.5

[a] Cross-references to variables defined in other modules.
All variables vary by firm and are based on 4-year averages (2018–2021). Units: $OPEX_p$ and $CAPEX$ in tkr; Energy in MWh; Power in MW.

Table 15: Output variables: Data Envelopment Analysis

Description	Variable-ID
50.3 Derived efficiency measures	
Efficiency score (*Effektivitet*)[a]	50.3.1
Super-efficiency score (*Supereffektivitet*)[b]	50.3.2
Efficiency potential (*Effektiviseringspotential*)[c]	50.3.3
Applied efficiency potential (*Justerad potential*)[d]	50.3.4
50.4 Efficiency-adjusted costs	
OPEX Efficiency adjustment	50.4.1
CAPEX Efficiency adjustment	50.4.2
OPEX after efficiency adjustment	50.4.3
CAPEX after efficiency adjustment	50.4.4

[a] DEA efficiency score; 1.0 indicates full efficiency relative to frontier.
[b] Super-efficiency allows scores above 1.0 for frontier firms.
[c] Raw efficiency potential before applying caps and bounds.
[d] Efficiency potential bounded between 0.08 and 0.30.
All variables vary by firm and are based on 4-year averages (2018–2021). Efficiency-adjusted costs in kr.

6 Model outputs

Here write more concisely about outputs

7 Add-on modules

7.1 Benchmarking module

Description of the Benchmarking module to be added here.

References

Appendix A: Additional tables and figures

Appendix B: